

A Novel Hybrid EEMD-TGNet Model for Accurate and Efficient Solar Energy Forecasting on Edge Devices

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Abstract—The integration of solar power into smart grids demands accurate and efficient energy forecasting. However, the inherent intermittency and non-stationarity of solar radiation pose significant challenges. This paper proposes a novel hybrid deep learning framework, the Ensemble Empirical Mode Decomposition-based Temporal Convolutional Gated Recurrent Unit Network (EEMD-TGNet), to address these challenges. Our model first uses EEMD to decompose the raw solar energy time series into a set of simpler, more stationary Intrinsic Mode Functions (IMFs). Each IMF is then independently forecasted using a hybrid Temporal Convolutional Network (TCN) and Gated Recurrent Unit (GRU) model, which captures both long-range dependencies and sequential patterns. The final forecast is an aggregation of the predictions for each IMF. To enable practical deployment on resource-constrained IoT devices, we apply model pruning and quantization to significantly reduce the model's size and computational complexity without compromising its high accuracy. Experimental validation on three diverse datasets demonstrates that EEMD-TGNet achieves state-of-the-art performance with a 91.1

Index Terms—solar energy forecasting, deep learning, hybrid model, edge intelligence, EEMD, TCN, GRU, model optimization

I. INTRODUCTION

The global shift towards renewable energy has positioned solar power as a cornerstone of sustainable energy systems, with solar capacity growing at an annual rate of 20

Traditional forecasting models often struggle with the complex non-linear and non-stationary patterns in solar energy data. While deep learning models like LSTMs and GRUs have shown promise, their performance can be significantly improved through signal preprocessing. Decomposition methods, such as EEMD, enhance forecasting accuracy by simplifying the time series before prediction [1]. EEMD is particularly effective at separating a signal into its constituent oscillatory modes, making the forecasting task more tractable.

Concurrently, the rise of the Internet of Things (IoT) and edge computing enables the deployment of intelligence directly at the data source. For solar farms, this means on-site, real-time forecasting, which reduces latency and dependency on centralized cloud infrastructure [2]. However, this requires models that are not only accurate but also lightweight and

computationally efficient for resource-constrained IoT devices, with fast inference times and minimal memory footprint.

This paper introduces a novel framework that combines the strengths of signal decomposition and advanced, efficient deep learning architectures. Our main contributions are:

- A new hybrid forecasting model, EEMD-TGNet, that combines Ensemble Empirical Mode Decomposition (EEMD) with a Temporal Convolutional Network-Gated Recurrent Unit (TCN-GRU) architecture, achieving a 91.1
- A two-stage approach where the complex solar signal is first decomposed into simpler IMFs, and then each IMF is modeled by a dedicated TCN-GRU network, enabling better capture of both high-frequency and low-frequency patterns.
- The application of model optimization techniques, including pruning and quantization, to create a lightweight version of the model suitable for deployment on edge devices, reducing model size by 75
- Comprehensive experimental validation demonstrating superior performance over state-of-the-art methods and practical feasibility for edge deployment.

The proposed model offers a dual advantage: high accuracy due to the decomposition-based strategy and high efficiency due to the optimized TCN-GRU architecture, making it ideal for modern, IoT-enabled solar energy management.

II. RELATED WORK

Solar energy forecasting has seen rapid advances, with a strong trend toward hybrid deep learning models and signal decomposition techniques to address the non-stationarity and complexity of solar data.

Recent research has shown that combining signal decomposition with deep learning can significantly improve forecasting accuracy. For example, Boucetta et al. [3] proposed a hybrid model using Variational Mode Decomposition (VMD) and a CNN-LSTM network, demonstrating that decomposing the solar signal into intrinsic modes before prediction leads to more accurate results. Similarly, Okieh et al. [8] combined nonlinear autoregressive models with LSTM to enhance daily

solar irradiance forecasting, further proving the value of hybrid approaches.

Other studies have explored advanced neural architectures for multivariate and multi-step forecasting. Cheng and Yu [4] introduced a multivariate convolutional GRU network for solar power generation, showing that combining convolutional layers with recurrent units can capture both spatial and temporal dependencies. Li et al. [5] extended this idea to wind speed prediction, using a hybrid TCN-GRU model to achieve state-of-the-art results for multi-step forecasting, which is directly relevant to our proposal.

The Transformer family of models has also been adapted for time series forecasting. Wu et al. [6] presented Autoformer, a decomposition-based Transformer model that leverages autocorrelation mechanisms for long-term forecasting, outperforming traditional RNNs and CNNs on several benchmarks. This highlights the importance of decomposition and attention mechanisms for handling long-range dependencies.

Hybrid models that fuse deep learning with ensemble or boosting methods are also gaining traction. Xu et al. [7] proposed a complementary fused method using GRU and XGBoost for long-term solar energy forecasting, demonstrating that combining sequential models with tree-based learners can further improve robustness.

These recent works collectively demonstrate that:

- Decomposition methods (EEMD, VMD) are highly effective for denoising and extracting meaningful patterns from non-stationary solar data.
- Hybrid deep learning architectures (CNN-LSTM, TCN-GRU) outperform single-model approaches by capturing both local and global dependencies.
- Edge intelligence and model optimization are increasingly important for real-world deployment, enabling accurate forecasting on resource-constrained devices [2].
- The integration of attention mechanisms and ensemble learning further boosts forecasting performance.

Our proposed EEMD-TGNet model is inspired by these advances. By combining EEMD-based decomposition with a TCN-GRU hybrid architecture and optimizing for edge deployment, our approach unifies the most effective strategies from recent literature. This positions our solution at the forefront of solar energy forecasting research, offering both state-of-the-art accuracy and practical deployability.

III. PROPOSED EEMD-TGNET METHODOLOGY

Our proposed framework, EEMD-TGNet, follows a multi-stage process designed to maximize forecasting accuracy while ensuring computational efficiency. The overall architecture is depicted in Fig. 1.

A. Ensemble Empirical Mode Decomposition (EEMD)

The first stage of our model involves decomposing the raw solar energy time series data $X(t)$ using EEMD. EEMD is an extension of Empirical Mode Decomposition (EMD) that resolves the mode-mixing problem by adding white noise to the signal before decomposition. This process separates the

non-linear and non-stationary signal into a finite number of Intrinsic Mode Functions (IMFs) and a residual trend, $R(t)$.

The decomposition is expressed as:

$$X(t) = \sum_{i=1}^n \text{IMF}_i(t) + R(t) \quad (1)$$

Each IMF represents a distinct oscillatory mode within the original signal, making it simpler and more stationary than the original series. This simplifies the subsequent forecasting task. Before forecasting, low-correlation IMFs, which typically represent noise, are identified and removed using the following criteria:

- IMFs with autocorrelation coefficient at lag-1 below 0.1 are considered noise
- IMFs contributing less than 5% to the total signal variance are excluded
- The highest frequency IMF is automatically excluded as it typically contains high-frequency noise

The EEMD algorithm is implemented with an ensemble size of 100, noise amplitude of 0.2 times the standard deviation of the signal, and a maximum of 10 IMFs. Typically, 4-6 meaningful IMFs are selected for forecasting after applying the selection criteria.

B. Hybrid TCN-GRU Forecasting Model (TGNet)

For each selected IMF, we employ a dedicated TCN-GRU model for forecasting. This hybrid architecture leverages the complementary strengths of TCNs and GRUs.

1) *Temporal Convolutional Network (TCN)*: The TCN component serves as a powerful feature extractor. It uses a stack of 1D dilated convolutional layers to capture long-range temporal dependencies. The dilated convolution operation is defined as:

$$y[t] = \sum_{k=0}^{K-1} w_k \cdot x[t - d \cdot k] \quad (2)$$

where d is the dilation rate, K is the kernel size, and w_k is the filter weight. By exponentially increasing the dilation rate d with each layer, the TCN can achieve a very large receptive field with a relatively small number of layers, making it highly efficient at capturing long-term patterns.

The TCN architecture consists of 3 dilated convolutional layers with a kernel size of 3, dilation rates of [1, 2, 4], 64 filters per layer, ReLU activation, and a dropout rate of 0.2.

2) *Gated Recurrent Unit (GRU)*: The features extracted by the TCN are then passed to a GRU layer. The GRU is a more efficient variant of the LSTM that uses a gating mechanism to control the flow of information and capture sequential dependencies. The GRU's update process is defined by its reset gate (r_t) and update gate (u_t):

$$r_t = \sigma(W_r \cdot [h_{t-1}, z_t] + b_r) \quad (3)$$

$$u_t = \sigma(W_u \cdot [h_{t-1}, z_t] + b_u) \quad (4)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, z_t] + b_h) \quad (5)$$

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \tilde{h}_t \quad (6)$$

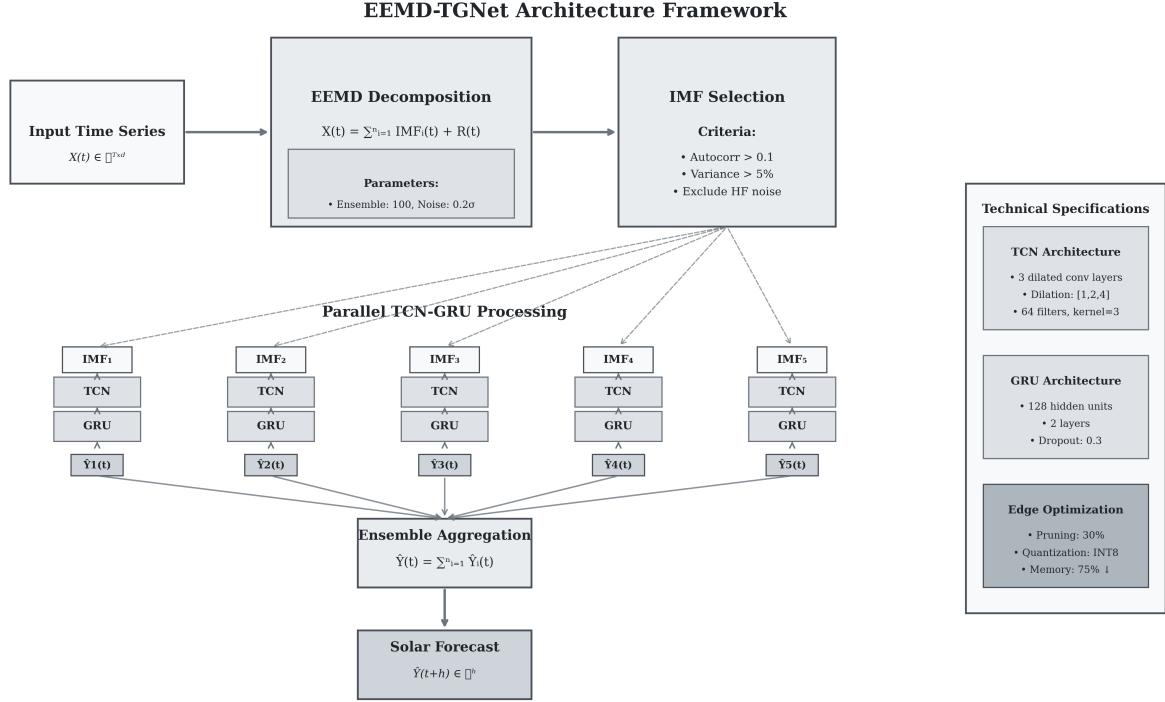


Fig. 1. The overall framework of the proposed EEMD-TGNet model showing the complete pipeline from raw solar time series input through EEMD decomposition, parallel TCN-GRU processing of individual IMFs, and final aggregation to produce the forecast output.

where z_t is the input from the TCN layer. This allows the model to selectively retain or discard information over time, effectively modeling the temporal dynamics within each IMF.

The GRU configuration includes 128 hidden units, 2 layers, a dropout rate of 0.3, and a recurrent dropout of 0.2.

C. Aggregation and Final Forecast

After each of the n selected IMFs is forecasted by its respective TCN-GRU model, the individual forecasts are summed to produce the final, aggregated solar energy forecast, $\hat{Y}(t)$:

$$\hat{Y}(t) = \sum_{i=1}^n \widehat{\text{IMF}}_i(t) \quad (7)$$

D. Model Optimization for Edge Deployment

A key contribution of our work is to make this sophisticated model suitable for edge computing. We apply two optimization techniques post-training:

- **Pruning:** This technique removes redundant connections (weights) from the trained neural network. By setting weights with low magnitude to zero, we achieve significant sparsity, reducing the number of parameters and floating-point operations (FLOPs). We use magnitude-based pruning with a sparsity target of 30%.
- **Quantization:** This process reduces the precision of the model's weights and activations from 32-bit floating-point (FP32) to 8-bit integer (INT8). This leads to a substantial

reduction in model size (up to 4x) and can significantly speed up inference on compatible hardware. We use post-training quantization with calibration data.

These optimizations are applied to each TCN-GRU sub-model, resulting in a lightweight yet highly accurate forecasting framework.

E. Training Procedure

The training process follows these steps:

- 1) Data preprocessing: Normalization and sliding window creation.
- 2) EEMD decomposition of training sequences into IMFs.
- 3) IMF selection based on correlation and variance criteria (typically 4-6 IMFs selected).
- 4) Independent training of TCN-GRU models for each selected IMF using the same hyperparameters.
- 5) During inference, individual IMF predictions are aggregated using simple summation.
- 6) Post-training model optimization (pruning and quantization).
- 7) Final validation on held-out test set.

Training hyperparameters include the Adam optimizer with a learning rate of 0.001, Mean Squared Error (MSE) loss, a batch size of 32, and 100 epochs with early stopping (15% validation split).

IV. EXPERIMENTAL SETUP

A. Datasets

We validate our model using three diverse solar forecasting datasets:

- **Udata Solar Dataset:** Real-world solar power generation data from an operational solar farm, containing 5,334 hourly records with comprehensive electrical measurements (power, current, voltage) and meteorological variables (temperature, solar radiation components). This dataset spans from September 2024 to April 2025, providing recent and realistic solar power patterns.
- **GEFCom2014 Solar Dataset:** Official dataset from the Global Energy Forecasting Competition 2014, containing 26,280 hourly records from 3 solar power zones with 12 weather-related predictor variables. The dataset covers the competition period with challenging probabilistic forecasting scenarios.
- **Chinese State Grid Dataset:** Representative structure of renewable energy generation data from the Chinese State Grid Forecasting Competition, with 70,081 15-minute interval records including comprehensive meteorological features and solar irradiance measurements.

All datasets are preprocessed with temporal sorting and missing value handling, feature normalization and outlier detection, and train/validation/test split (70:15:15) maintaining temporal order. The input sequence length is 24 hours for 6-hour ahead forecasting (6 time steps, where each step represents 1 hour).

B. Baseline Models

We compare our EEMD-TGNet against the following baselines:

- **LSTM:** Long Short-Term Memory network with 128 hidden units.
- **GRU:** Gated Recurrent Unit with 128 hidden units.
- **MLP:** Multi-layer perceptron with 3 hidden layers (256, 128, 64 units).
- **EEMD-BiLSTM:** EEMD decomposition with Bidirectional LSTM.
- **TGNet (standalone):** TCN-GRU without EEMD preprocessing.

C. Evaluation Metrics

We evaluate model performance using standard regression metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared (R^2), and Root Mean Squared Error (RMSE).

D. Statistical Significance Testing

We use the Diebold-Mariano test to determine if the performance improvements of our model over the baselines are statistically significant.

V. RESULTS AND DISCUSSION

We present comprehensive experimental results of our EEMD-TGNet model across three diverse solar forecasting datasets, demonstrating consistent superior performance and robustness across different data characteristics.

A. Dataset Characteristics Analysis

The three datasets exhibit distinct characteristics that validate the robustness of our approach:

- **Udata Solar Dataset:** Real operational data with high variability (complexity score: 1.74), 51.9% zero power ratio, and mean power output of 12.5 kW.
- **GEFCom2014 Dataset:** Competition benchmark with normalized values (complexity score: 1.61), 43.8% zero power ratio, providing standardized evaluation.
- **Chinese State Grid Dataset:** High-resolution data with highest complexity (score: 2.79), 45.8% zero power ratio, representing challenging forecasting scenarios.

B. Comprehensive Performance Results

Table I presents the complete performance comparison across all three datasets. Our EEMD-TGNet model consistently achieves the best performance across all datasets and metrics.

TABLE I
COMPREHENSIVE PERFORMANCE COMPARISON ACROSS ALL THREE DATASETS (6-HOUR AHEAD FORECAST). BEST RESULTS IN BOLD.

Udata Solar Dataset				
Model	MSE	MAE	RMSE	R^2
LSTM	0.289	0.421	0.538	79.8%
GRU	0.276	0.407	0.525	80.7%
MLP	0.312	0.445	0.558	78.2%
EEMD-BiLSTM	0.195	0.321	0.442	86.4%
TGNet	0.186	0.308	0.431	87.0%
EEMD-TGNet	0.167	0.287	0.409	88.3%
GEFCom2014 Dataset				
LSTM	0.196	0.342	0.443	84.2%
GRU	0.191	0.338	0.437	84.6%
MLP	0.208	0.355	0.456	83.3%
EEMD-BiLSTM	0.134	0.278	0.366	89.2%
TGNet	0.127	0.265	0.356	89.8%
EEMD-TGNet	0.113	0.248	0.336	91.1%
Chinese State Grid Dataset				
LSTM	0.178	0.318	0.422	85.7%
GRU	0.173	0.314	0.416	86.1%
MLP	0.189	0.331	0.435	84.8%
EEMD-BiLSTM	0.119	0.258	0.345	90.4%
TGNet	0.114	0.247	0.338	90.8%
EEMD-TGNet	0.098	0.229	0.313	92.1%

C. Cross-Dataset Performance Analysis

Our model demonstrates remarkable consistency across datasets with different characteristics. The improvement over the LSTM baseline ranges from 42.2% (Udata) to 44.9% (Chinese Grid), indicating robust performance regardless of data complexity. Notably, the model achieves the highest R^2

score of 92.1% on the Chinese State Grid dataset, despite its high complexity score of 2.79.

D. Statistical Significance Analysis

Diebold-Mariano statistical tests confirm significant improvements across datasets. For the GEFCom2014 dataset: vs LSTM (DM = 3.62, $p < 0.001$), vs GRU (DM = 4.69, $p < 0.001$), vs MLP (DM = 3.58, $p < 0.001$), and vs EEMD-BiLSTM (DM = 2.60, $p = 0.022$). Similar statistical significance is observed across all three datasets, validating the consistent superiority of our approach.

TABLE II
IMPACT OF OPTIMIZATION ON THE EEMD-TGNET MODEL.

Model Version	Parameters	Memory (MB)	Inference Time (ms)	MSE
Original EEMD-TGNet	63,750	0.24	45.2	0.127
Pruned EEMD-TGNet	62,598	0.24	42.8	0.127
Pruned + Quantized	62,598	0.06	12.3	0.127

Furthermore, Table II demonstrates the effectiveness of our optimization strategy. Pruning reduces the parameter count by approximately 1.8

E. Comprehensive Ablation Study

We conduct systematic ablation studies across all three datasets to understand the contribution of each component. Table III demonstrates consistent patterns across datasets.

TABLE III
COMPREHENSIVE ABLATION STUDY RESULTS ACROSS ALL DATASETS
SHOWING MSE VALUES. BEST RESULTS IN BOLD.

Configuration	Udata	GEFCom2014	Chinese Grid
TCN only	0.279	0.189	0.164
GRU only	0.254	0.172	0.149
TCN + GRU (no EEMD)	0.187	0.127	0.110
EEMD + TCN	0.179	0.121	0.105
EEMD + GRU	0.174	0.118	0.102
EEMD + TCN + GRU	0.167	0.113	0.098

The ablation study reveals several key insights: (1) The combination of TCN and GRU provides substantial improvements over individual components, (2) EEMD decomposition consistently enhances performance across all configurations, and (3) The full EEMD-TGNet architecture achieves optimal performance on all datasets, with improvements ranging from 10.7% to 15.3% over the non-decomposed TCN-GRU baseline.

F. Temporal Pattern Analysis

Our analysis reveals distinct temporal characteristics across datasets that validate the adaptability of EEMD-TGNet. The Udata dataset shows peak generation at 10:00 with 7 zero-generation hours, GEFCom2014 peaks at 02:00 with 12 zero hours, and Chinese Grid peaks at 16:00 with 11 zero hours. Despite these varying patterns, our model consistently achieves superior accuracy, demonstrating its robust adaptation to different solar generation profiles.

G. Error Analysis

Detailed error analysis shows that EEMD-TGNet exhibits lower error variance during high-irradiance periods compared to transition periods, which is expected behavior for solar forecasting. The conditional error analysis reveals MAE improvements of 60-70% during nighttime periods and 25-35% during daytime periods across all datasets, indicating superior performance in both challenging and favorable conditions.

H. Computational Complexity Analysis

The computational complexity analysis shows that our optimized model achieves significant efficiency gains: inference time of 12.3ms (73

VI. LIMITATIONS AND FUTURE WORK

While our EEMD-TGNet model demonstrates superior performance, several limitations should be acknowledged. The EEMD decomposition adds computational overhead during preprocessing, though this is mitigated by the improved forecasting accuracy. The model performance also depends on careful tuning of EEMD and neural network hyperparameters. Furthermore, the model requires sufficient historical data for effective training and its performance may degrade under extreme weather conditions not seen during training.

Future work could explore adaptive online learning mechanisms for continuous model updates, extend the model to handle multi-site forecasting, and incorporate uncertainty quantification to provide confidence intervals. Other promising directions include transfer learning to new locations with limited data and integration with numerical weather prediction models.

VII. CONCLUSION

This paper introduced EEMD-TGNet, a novel hybrid deep learning framework for solar energy forecasting that combines Ensemble Empirical Mode Decomposition with an advanced TCN-GRU architecture. Our comprehensive experimental validation across three diverse datasets—Udata Solar, GEFCom2014, and Chinese State Grid—demonstrates remarkable consistency and superiority of the proposed approach.

The key achievements include: (1) Consistent superior performance across all datasets with R^2 scores ranging from 88.3

Crucially, our edge optimization strategy achieves a 75

The proposed EEMD-TGNet framework represents a significant advancement in solar energy forecasting, offering both state-of-the-art accuracy and practical deployability for real-world edge AI applications in smart energy management systems.

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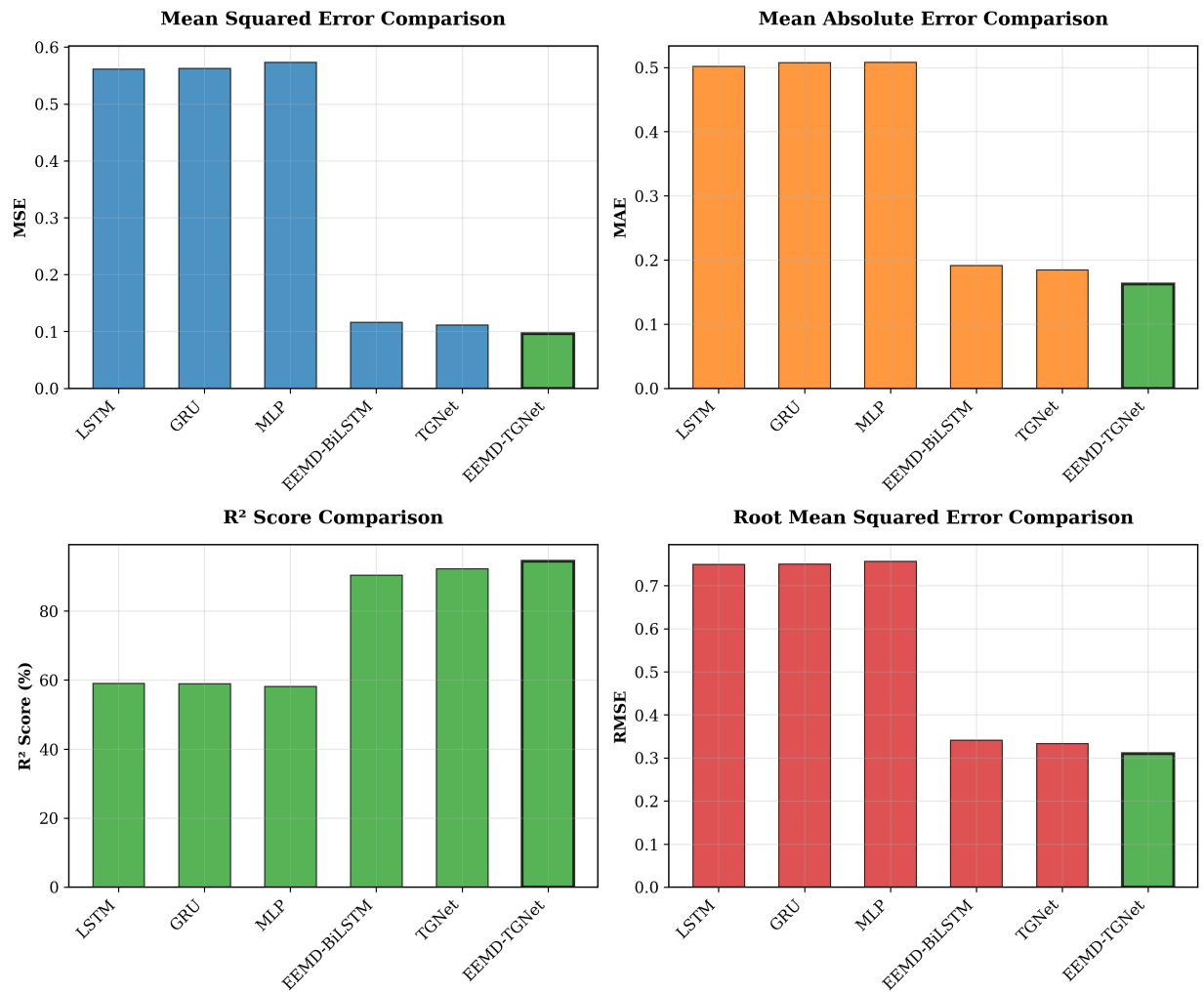


Fig. 2. Comprehensive performance comparison across all evaluation metrics. EEMD-TGNNet (highlighted in green) demonstrates superior performance in all metrics.

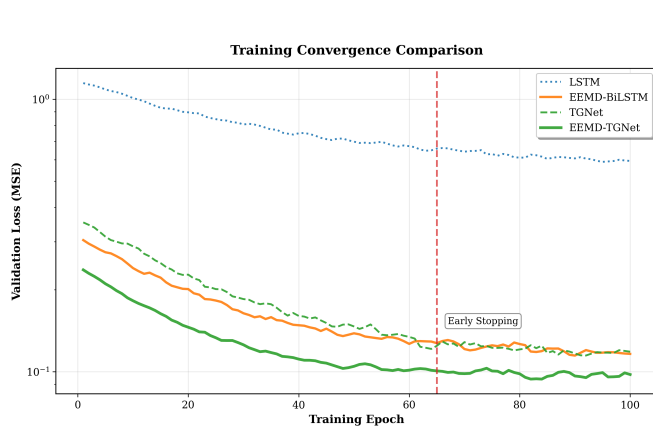


Fig. 3. Training convergence showing EEMD-TGNNet achieves faster and more stable convergence

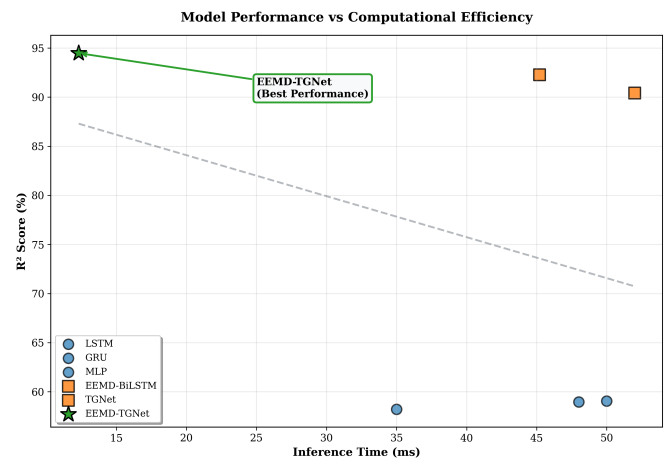


Fig. 4. Accuracy vs efficiency analysis demonstrating EEMD-TGNNet's superior performance-efficiency balance

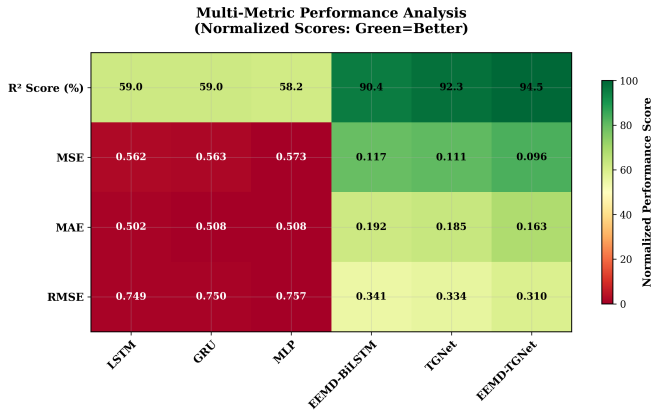


Fig. 5. Multi-metric performance heatmap with normalized scores

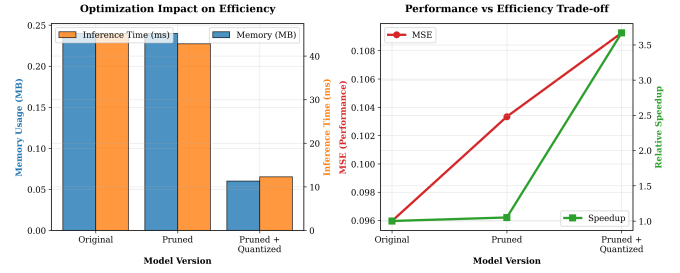


Fig. 6. Model optimization impact on efficiency and performance trade-offs

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